

Mathematics Reference: Problems on Ages

Educational Series

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1 Core Concept

Problems on ages are essentially linear equations in one or two variables. The primary challenge is translating verbal statements regarding "past," "present," and "future" into mathematical expressions.

2 Translation Table

Let the current age of a person be x .

Expression in Words	Mathematical Expression
Age n years ago	$x - n$
Age n years from now (hence)	$x + n$
n times the current age	nx
If age n years ago was y	$x - n = y$
Age will be y after n years	$x + n = y$

3 Important Properties

- **Constant Difference:** The difference between the ages of two people remains constant throughout their lives. If A is 5 years older than B today, A will still be 5 years older than B twenty years from now.
- **Ratio Shifts:** While the age difference is constant, the *ratio* of their ages changes over time.
- **Positive Constraint:** Age is always a positive quantity ($x > 0$).

4 Step-by-Step Strategy

1. **Identify the Unknown:** Usually, it is best to let the *present age* be x .
2. **Establish Relations:** Create expressions for their ages in the past or future as mentioned in the problem.
3. **Form the Equation:** Use the condition given in the problem (e.g., "the sum of their ages is 50") to link the expressions.
4. **Solve for x :** Calculate the value and ensure it answers what was specifically asked (present age vs. age in 5 years).

5 Solved Example

Problem: The ratio of the ages of Father and Son at present is 4 : 1. After 5 years, the ratio will become 3 : 1. Find their present ages.

Solution:

1. Let the present age of the Son be x .
2. Then the present age of the Father is $4x$.

3. **After 5 years:**

- Son's age = $x + 5$
- Father's age = $4x + 5$

4. **Forming the equation:**

$$\frac{4x + 5}{x + 5} = \frac{3}{1}$$

5. **Solving:**

$$4x + 5 = 3(x + 5)$$

$$4x + 5 = 3x + 15$$

$$x = 10$$

6. **Conclusion:** The Son's present age is **10 years** and the Father's present age is $4 \times 10 =$ **40 years**.